Primality Testing

Module I

Primality testing

- Given an integer n, can we tell if n is prime?

Fermat Test for primality

- If n is prime number then for any a we have

$$a^{n-1} \equiv 1 (mod\ n)\ 1 \leq a < n$$

1. If $a^{n-1} \equiv 1(mod\ n)$, declare n a probable prime, and optionally repeat the test a few more times.
2. If $a^{n-1} \not\equiv 1(mod\ n)$, declare n composite, and stop.

Example: Test with fermat's little theorem

- Check if n is a prime number n=5. check a value with 1, 2, 3, and 4. More number of values checking to get more accuracy.

Problems with fermat's theorem

- If a number fails fermat's test then it is certainly composite.
- Lets check if n=4 a prime number using Fermat's little theorem.

Problems with fermat's theorem

- A number may pass the test but not be prime.
- Lets check if 561 is a prime number as per fermat's little theorem.